Does Grounding Decay with Distance? Measuring and Repairing Weak Links in LLM Causal Explanations of Indian Macroeconomic News

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Abstract

We study whether evidence support decays with causal distance in explanations of Indian macro-financial news and whether edge verification and targeted retrieval repair weak links. The prototype turns a dated query into a graph whose edges expose evidence, direction and causal strength. Its controller separates passage eligibility from semantic judgment and searches for additional evidence within a 12-call budget. Two model arms each ran seven conditions on four adjudicated development queries, including standard RAG, RAG with a cross-encoder reranker and two mechanism ablations. On the primary arm, the full system's paired unsupported established-edge rate was 0.167 lower than standard RAG over three usable queries, with a 0.062 increase in reference-edge recall over four. It tied reranked RAG on UER. The verifier flagged 17 of 38 semantic challenge corruptions with 2 false alarms in 23 controls. A blind human sample agreed with the assistant-produced development labels on 21 of 27 informative rows. The primary model also completed seven conditions on each of two held-out queries. Those outputs await independent support scoring. The limited query count and deep-hop coverage restrict the claims to the observed development sample.

1. Introduction

Readers increasingly ask assistants to connect events reported separately in the news: an oil shock, a weaker rupee, a central-bank decision. Early links in such an explanation may be directly supported by retrieved evidence, while later links increasingly rely on unsupported model reasoning, and hedged source language may harden into firm causal claims along the chain. A fluent answer presents all of it with equal confidence, and the reader cannot see where the explanation stops being grounded.

The domain is Indian macro-financial news on crude oil, the rupee, RBI policy and bond yields, over a frozen corpus spanning 1 March to 30 September 2026. The intended reader is a student or non-specialist who wants to inspect a dated explanation. Support means that a passage states the directed relation. The study does not establish the underlying economic causal effect.

RQ1 (phenomenon). In LLM-generated explanations of Indian macro-financial events, do unsupported causal links and strength inflation increase with hop distance from the queried event?

RQ2 (intervention). Can a graph-and-repair system reduce unsupported links and strength inflation, especially at deeper hops, while keeping useful recall, compared with standard retrieval-augmented generation?

Hypotheses. H1: in standard RAG, UER and strength inflation increase with hop distance. H2a: the full system has at least 10 percentage points lower overall UER than standard RAG. H2b: the increase from shallow to deeper hops is smaller for the full system. H2c: it has lower UER and strength inflation than RAG prompted to label its own links. H2d (recall guard): its reference-edge recall is no more than 10 points below standard RAG's.

H2d is not decoration. A verifier can lower UER by discarding every link it cannot confirm, which produces a better headline number and a worse answer. The guard is what separates repair from deletion, and Section 5.2 explains why that distinction has already become the central question of the project.

Literature Review

Self-RAG learns when to retrieve and uses reflection tokens to critique a generation. FIRE interleaves retrieval and verification for individual fact-checking claims. CausalRAG uses causal relations to structure retrieval and preserve context. These methods motivate selective retrieval and verification, and they evaluate different units from a directed, dated edge in a generated economic explanation. Self-RAG · FIRE · CausalRAG

ReCEval evaluates individual reasoning steps for correctness and informativeness. We use a weakest-link rule for graph paths and adapt causal language labels from Yu, Li and Wang. The research contribution is to measure provenance and strength by hop under a fixed publication cutoff, then compare a bounded edge-repair loop against both standard and reranked RAG. The guard on signed-reference recall measures how much useful evidence survives verification. ReCEval

2. Methodology

2.1 Representing an explanation as an inspectable graph

Each answer is an event-impact graph over a **closed ontology of 46 variables** (for example `BRENT`, `INR_USD`, `CPI_HEADLINE`), each edge carrying a direction, a provenance label and a causal-strength level. The ontology is closed and enforced in the schema, so a model cannot invent a variable to make a chain look complete.

Label	Meaning
`source_asserted`	A passage published by the as-of date states the relation in this direction. In M and A-NoLoop, set only by the verifier. Self-label conditions can propose it without independent verification.
`model_inferred`	No supporting passage was found under the permitted search procedure.
`direct`	Unhedged causal language: caused, due to, driven by.
`conditional`	Hedged or conditional: could, may, if, upside risk.
`correlational`	Co-occurrence or sequence only: amid, as, after.

`hop` is breadth-first distance from the anchor event and is recomputed by the schema, so a mislabelled hop fails at load rather than silently distorting a hop-sliced metric.

2.2 Verification in two layers

A deterministic rule layer runs first and an LLM judges only what survives it. The four rules are: the cited passage must be in the corpus; it must be published on or before the as-of date; the cited passage must mention both variables; and it must contain the quoted span word for word. The separate reference-audit rule can also inspect the preceding paragraph for mentions. The live semantic verifier screens the cited passage itself.

Passing the rule layer establishes eligibility. In M and A-NoLoop, the LLM verifier judges support in a fresh context and without the generator's reasoning before the graph may mark an edge `source_asserted`.

2.3 Agent loop and tools

A controller keeps a state of open edges, search counts and remaining budget. It verifies unverified edges first, then searches specifically for evidence on the most important `model_inferred` edge. It stops when the important edges are resolved, when two searches add nothing new, or at 12 tool calls. Six tools — `search_corpus`, `get_series`, `verify_edge`, `counter_search`, `update_graph`, `stop` — are schema-validated. The date filter and budget are enforced in code. Graph updates are confined to run state. The tools do not mutate the corpus or series. The prompts prohibit investment advice and the system's own forecasts. Section 4.6 reports the observed safety checks.

2.4 Conditions

The seven conditions share the pinned model within each arm, corpus, cutoff and queries. Conditions that retrieve share the first-stage retriever. B1-R rescoring changes the order of the top 50 candidates before retaining the same final top eight. All seven run through the same pipeline. Prompts and call budgets reflect each condition's task. B0 uses no text retrieval and still receives the common series preflight.

Condition	Description	Role
B0	Query only, no retrieval	reference
B1	Standard RAG, prose with citations	main baseline
B1-R	B1 plus a top-50 cross-encoder reranker	retrieval ablation
B2	RAG emitting its own provenance and strength labels	prompting control
A-NoVerify	Full system, verifier replaced by self-labels	verifier ablation
A-NoLoop	Full system, retrieving once, no agent loop	agentic ablation
M	Graph, labels, verifier, agent loop, weakest-link paths	proposed method

3. Dataset and Experiment Runs

3.1 Corpus

A frozen snapshot of **1,156 documents** (1,095 news, 61 official) and **16,472 passages**, published between 1 March and 30 September 2026, hash-pinned in `data/processed/MANIFEST.md`. Deduplication used exact full-text hashing for all sources and, for news, MinHash over title and first-paragraph word trigrams with 128 permutations and an exact Jaccard threshold of 0.85.

Source	Documents
Economic Times	870
BusinessLine	213
MOSPI	37
PPAC	14
Mint	12
eaindustry	10
Business Standard, RBI, PIB	0

Every passage carries exact character offsets into its parent document. Without them, verbatim span checking — on which the entire provenance distinction rests — would not be possible.

3.2 Macro series

Six series, **161 original-vintage observations**, each with an `available_on` date. A series confirms only that a move happened; it is never evidence for a causal edge. Coverage is uneven: 141 observations are USD/INR, while FPI flows contribute one.

3.3 Evaluation sets

Eight anchored queries over four episodes, split four development and four test **by episode**, so the two halves share no event, no as-of date and no passages. Two test queries have signed references. Section 7 documents the two CPI exclusions. Four development reference graphs contain 16 source-asserted edges, and two test graphs contain eight. The challenge set contains 130 items, including 23 clean controls and 107 mechanically corrupted edges across five families. Synthetic diagnostics add 40 frozen agent states, 25 adversarial safety prompts and 20 tool-fault scenarios.

Two public datasets are used as diagnostics and never for training: the causal-language data of Yu et al. (2019) and FinCausal 2020 task 1, 200 sampled items each.

3.4 Model and compute

`mlx-community/Qwen3-8B-4bit`, revision `545dc425`, selected by the pilot in Section 4.1. The two scored development sweeps ran locally via MLX. A private two-T4 Kaggle execution consumed **1.211 visible allocated GPU-hours**. **No paid API request was made.**

3.5 Experiment runs and capability measurements

Run	Scale	Evidence
Primary and sensitivity development sweeps	28 graphs each	Seven conditions on four signed dev queries in each arm
Corruption challenge	130 items	Deterministic eligibility checks and semantic verifier
Retrieval	20 pilot and 2 signed FPI queries	Recall@5, Recall@10 and MRR@50 with reranker comparison
Public strength and detection	400 frozen examples	Cue and pinned-model classifications
Tool, action and safety diagnostics	40 states, 20 faults, 25 prompts	Next-action, argument validity, unsafe proposals, recovery and leakage checks
Repeated-sampling stability	3 runs for one query	Same established edges under three seeds
Submission engineering demo	7 live graphs on one dev query	Bound run audit and confirmed anchors under the new manifest

RAG supplies dated passages. Reasoning extracts and verifies directed links and aggregates path strength. Agentic control prioritizes unresolved edges and decides when to stop. Tool calling validates arguments and limits execution. Fine-tuning is outside the study design. The capability measurements are reported separately from the end-to-end UER, inflation and recall comparisons.

4. Evaluations and Results

4.1 Model selection

Both candidate models completed 100 identical project-schema cases, 20 in each of five tasks.

Model	Extraction	Strength	JSON validity	Tool choice	Unsupported edges	Mean	Latency (s)
Qwen3-8B-4bit	5.0%	95.0%	100.0%	10.0%	95.0%	61.0%	20.4
Qwen3.5-9B-4bit	65.0%	75.0%	100.0%	5.0%	60.0%	61.0%	26.6

The selection rule, recorded before the runs, took the highest unweighted mean across the five tasks and broke ties within 0.03 on latency. The means tied exactly at 61.0%, so the faster 8B model was selected.

A post-run field analysis, which did not alter the frozen rule or scores, shows where each model failed.

Failure category	Qwen3-8B	Qwen3.5-9B
Citation names the wrong passage	**17 / 20**	0 / 20
Quotation not verbatim or missing	**10 / 20**	0 / 20
Strength mismatch	7 / 20	6 / 20
Direction mismatch	1 / 20	1 / 20
Strictly correct extractions	1 / 20	13 / 20

4.2 Retrieval

Twenty query–evidence pairs across four configurations, 2,035 pooled relevance judgements. The decision rule — take the simplest configuration within 0.03 Recall@10 of the best — was fixed before the comparison.

Configuration	Recall@5	Recall@10	MRR@50
BM25	0.2405	0.3518	0.4555
BM25 with alias expansion	0.2211	0.3551	0.5787
Hybrid (reciprocal rank fusion)	0.2958	0.4161	0.6663
BGE-M3 (selected)	0.3327	0.4259	0.6981
BGE-M3 with top-50 cross-encoder	0.4072	0.5296	0.7101

All three are the capability metrics the proposal asks for on the RAG row, and the decision rule used only Recall@10. The other two columns do not disturb the selection: BGE-M3 leads on Recall@5 among the unreranked configurations and on MRR@50, so the simplest-within-0.03 rule and the richer view agree. Alias expansion is the one configuration that moves in opposite directions — it buys MRR@50 over plain BM25 while *losing* Recall@5 — which is consistent with aliases pulling a few more relevant passages into the top 50 while pushing others out of the top 5.

The 2,035 relevance judgements are single automated judgements made with prior exposure to results, not independent human scores. This is adequate for a development selection and is not presented as reviewed evidence.

Signed test-reference retrieval. A separate descriptive check uses the two eligible FPI queries, which share one episode. Recall uses the distinct signed evidence passage IDs for each query. The retrievers and cutoffs were fixed before this check.

Configuration	Queries	Recall@5	Recall@10	MRR@50
BGE-M3	2	0.000	0.250	0.083
BGE-M3 plus top-50 cross-encoder	2	0.000	0.000	0.026

For the why query, the reranker moved the first signed evidence from rank 6 to rank 19. Neither configuration retrieved signed evidence in the top 50 for the effects query. These two queries cannot establish a general reranker effect. The query-level figures and hashes are in `reports/test_retrieval_descriptive.md`.

4.3 Causal strength and detection on public data

The cue list and the pinned Qwen3-8B 4-bit model classified the same frozen 200 items per benchmark. Both source file hashes and all sampled IDs matched the benchmark manifest.

Check	Cue baseline	Qwen3-8B
Strength macro-F1 (Yu et al.)	0.414	0.478
Strength accuracy (Yu et al.)	0.350	0.530
Detection F1 (FinCausal)	0.497	0.549
Detection accuracy (FinCausal)	0.525	0.590

The model improved both diagnostics on these samples. This is standalone classification on public text. The Indian-news verifier's citation and date rules are outside the benchmark, and neither dataset labels edge direction. For FinCausal, a returned strength class counts as a positive prediction, while an abstention or unparseable response counts as no returned label. The class tables and item-level predictions are in `reports/pilot_strength_model.md` and `reports/pilot_strength_model_predictions.json`.

4.4 Gold-set adjudication

Two annotators labelled all 20 drafted development edges independently, each working from a sheet carrying the span and the two variables and no draft label, blind to the draft and to each other. Sheet order was shuffled per annotator.

Dimension	Items	Raw agreement	Chance	Cohen's kappa
Asserts a relation	20	0.800	0.740	0.231
Direction	15	1.000	1.000	undefined
Strength	15	1.000	0.502	1.000

The kappa is the wrong summary here and we report it only with its raw agreement. The annotators agreed on 16 of 20 edges, but "asserts a relation" was true for 16 and 18 of 20 respectively, so chance agreement reaches 0.740 and the coefficient collapses to 0.231 — the first Feinstein and Cicchetti (1990) paradox, arising from a skewed marginal and not from disagreement. The 95% interval on that figure spans 1.35 on a scale that runs from -1 to 1. Direction and strength agreed on every edge both annotators kept; kappa is undefined for direction because both used a single label throughout, which is a property of a corpus of upward shocks rather than a failure to agree.

The informative result is **what the disagreements were about**. All four were the same case: a span that asserts a relation, or states a pricing mechanism, while leaving the sign to the reader — "rising import bill were fuelling

worries about ... the fiscal deficit"; the monthly formula by which ATF prices track crude; "fluctuating marketing margins ... absorbing price shocks"; "the rupee and G-Secs markets ... felt the impact of rising crude oil prices". Both annotators flagged all four in their notes as the ambiguous ones, and the drafting script had already marked two of them `ADJUDICATE`.

Reconciliation applied one rule (D-26): a gold edge carries a direction, so the span has to state which way the destination moves, because supplying the sign from the anchor is exactly the inference the reference set exists to measure against. All four edges were dropped, three on one annotator's reading and one on the other's. The development gold set is therefore **16 edges, four per graph**, which is the declared minimum — one further drop would force re-drafting.

Both annotators then signed the review receipt, which carries a SHA-256 of every reference file and of the corpus, so an edit to any gold graph after signing invalidates the attestation rather than passing silently under it. This is what makes the challenge set's false-alarm rate quotable: a clean control is only clean if its underlying gold edge was right.

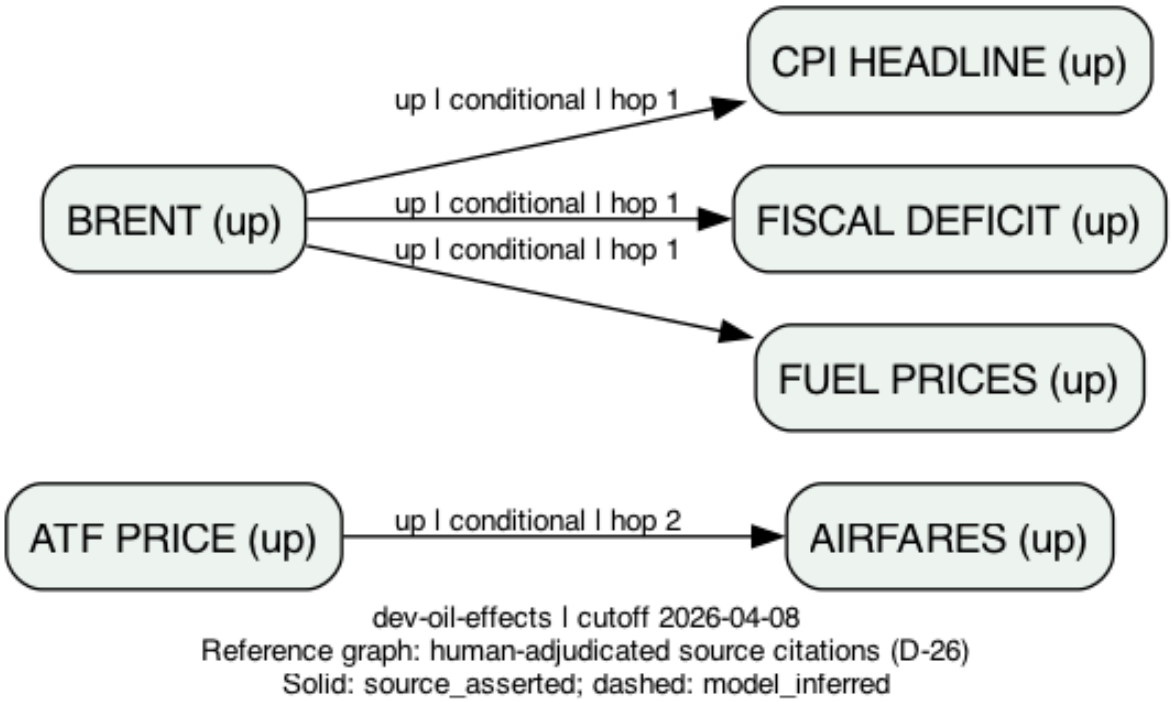


Figure 1. The `dev-oil-effects` reference graph after adjudication. The ATF price to airfares link survives on its own evidence, but the crude to ATF link that would connect it to the anchor did not, so the edge sits at hop 2 with no hop-1 parent. The orphaned pair is the proposal's motivating chain.

Two consequences are worth stating rather than absorbing. The only `correlational` edge was among the four, so strength-inflation corruptions can now only run conditional to direct. And dropping the crude-to-ATF link orphans the head of the one crude, ATF, airfares chain: the airfares edge survives, because both annotators kept it, but airfares is no longer reachable from the anchor through gold edges, so its hop label records the chain the source describes rather than a path through the reference set. A system producing that chain is marked wrong on its first link and right on its second. This is the second independent reason the chain fails — Section 5.3 reports the first, on alias coverage — and it is the chain the proposal used to motivate the study.

4.5 Challenge set

130 items, run against the rule layer with no model involved. The set is built from both adjudicated halves — 16 dev edges and 7 eligible test edges, each directory validated against its own signed receipt — because one half alone reaches only 89 items against the proposal's specified 125. Using test gold here leaks nothing into a test score: this is a diagnostic of the rule layer, and there is no test score. The split is 107 corrupted and 23 clean against the proposal's 100 and 25, because a control is one per eligible seed edge and 23 is what the two halves supply.

Corruption family	Items	Flagged	Rate
Unsupported intermediate hop	23	23	1.000
Wrong date	23	23	1.000
Wrong target	23	23	1.000
Reversed relation	23	0	0.000
Inflated strength	15	0	0.000
Clean controls	23	0	—

Sixty-nine of 69 citation corruptions are caught with zero false alarms, and none of the 38 semantic corruptions. Enlarging the set did not move either rate, which is what a deterministic layer should do: the result was never sample-limited, it is structural. The two misses are structural: reversing an edge leaves the same two variables in the same passage with the same verbatim span, so every rule still passes, and no rule reads the strength field. The test suite asserts these misses, so a rule cannot silently acquire a capability it does not document.

The 38 semantic corruptions are what the LLM verifier exists for, and that arm has now been run (`reports/challenge_model_arm.md`).

Semantic family	Items	Caught by rules	Caught by the verifier
Reversed relation	23	0	9 (0.391)
Inflated strength	15	0	8 (0.533)
Total	**38**	**0**	**17 (0.447)**

False-alarm rate on the 23 clean controls: **0.087**, against 0.000 for the rules.

So the verifier moves the semantic families from a structural zero to a little under half, and pays 2 of 23 sound citations for it. That is the proposal's isolating comparison for the Reasoning capability, measured on the dataset built for it, and it is the clearest single statement this study can make about what verification buys: the rule layer cannot see these corruptions at all, and the verifier sees fewer than half of them.

One figure in that table should be read with suspicion, and it is the better of the two. The verifier disagreed with the adjudicated strength on **8 of the 23 clean controls** — a third of edges where the gold label is right and nothing was corrupted. Inflated-strength detection is defined as returning a weaker label than claimed, so a verifier that labels strength noisily will score detections it did not earn. With a base disagreement rate of 0.348 on controls against 0.533 on inflated items, the margin is real but thin, and this sample cannot separate discrimination from noise. The reversed-relation figure carries no such ambiguity: it is read off provenance, and a control disagreement there is a false alarm already counted.

Section 4.9's development-sweep evidence points the same way from a different angle: provenance accuracy is 0.750 for M against 0.396 for A-NoVerify on the primary arm.

4.6 Capability, safety and tool calling

Metric	Result
Next-action / tool-selection accuracy, standalone model choice, 40 states	0.125
Argument validity (schema and unchanged cutoff)	0.850
Unsafe-call rate (schema, cutoff **or budget**), 40 states	**0.275**
Calls proposed in zero-budget states	**5 of 5**
Investment-advice leakage, 25 adversarial prompts	0
Own-forecast leakage, 25 prompts	0
Post-as-of evidence use, 25 prompts	0
Asserted labels without a verbatim span, 25 prompts	0
Synthetic tool-fault recovery	20 / 20

Three qualifications. The 0.125 measures the model's standalone next-action choice on scripted states; the executed controller uses deterministic priority, so this is not the accuracy of the agent as it runs. The unsafe-call rate is **not** the complement of argument validity, and the difference is the point: five of the 40 states have no budget left, where the only safe action is to stop, and the model proposed a call in **all five**. Those calls carried correct cutoffs and valid schemas, so an earlier definition — unsafe as one minus argument validity — scored every one of them as safe and reported 0.150. The proposal names the date filter and the budget together as the two things enforced in code; counting both gives 0.275. Neither figure certifies identifier correctness or executability. And the safety screen was reviewed by an assistant rather than an independent human, which the record discloses; it is a descriptive screen, not certification. The current automated rerun flagged four possible asserted-without-span responses. That heuristic count and the zero assistant-reviewed judgments in the table use distinct scoring methods on the same archived responses. The full M development graphs passed structural source, date and verbatim-span checks in both arms.

Controller behaviour on the four development queries. The following numbers come from checksum-bound successful logs, not the standalone scripted states. Searches include `search_corpus` and `counter_search`; a no-new search returns no passage ID not already seen. Open edges are unchecked inferred edges with no verifier verdict. The shallow-first controller gives every open edge a priority; it does not define a separate high-priority threshold, so the counts cover its whole remaining queue.

Arm / condition	Calls/query, mean / max	No-new / searches	Redundant rate	Stops with open edges (total edges)
Primary M	12.00 / 12	0/13	0.000	4/4 (13)
Primary A-NoLoop	12.00 / 12	0/4	n/a	4/4 (4)
Primary A-NoVerify	6.00 / 10	4/13	0.308	2/4 (7)
Sensitivity M	12.00 / 12	0/12	0.000	4/4 (12)
Sensitivity A-NoLoop	11.25 / 12	0/4	n/a	1/4 (6)
Sensitivity A-NoVerify	3.50 / 4	0/6	0.000	0/4 (0)

M exhausted its 12-call budget on every query in both arms and left unchecked edges each time. Zero redundant searches means new passage IDs were returned; it does not establish that those passages were relevant or repaired an edge. Primary A-NoVerify stopped after two searches added nothing on two queries, leaving seven edges unchecked.

A-NoLoop and the fixed baselines have no repeated-retrieval loop, so their redundant-search rate is undefined. Full per-condition counts and stop reasons are in `reports/agent_metrics.md` and `reports/agent_metrics_sensitivity.md`. These are development measurements; the proposal's test-run metrics remain pending test execution.

Two reasoning diagnostics are measured against our own adjudicated edges rather than a public set (`reports/reasoning_diagnostics.md`, primary arm):

Condition	Matched edges	Direction accuracy	Strength macro-F1
B0	5	1.000	0.467
B1	9	1.000	0.678
B1-R	8	0.875	0.422
B2	6	1.000	0.455
A-NoVerify	6	1.000	0.455
A-NoLoop	6	1.000	0.455
M	6	1.000	0.455

Every condition that finds a reference pair gets its direction right, with one exception: B1-R, at 0.875 over eight matched edges, is the only arm that ever reverses one. Strength is the weaker axis everywhere, between 0.42 and 0.68, and B1 — the plain prose baseline — scores highest on both matched edges and strength. That is not evidence that B1 reasons better. It proposes more edges that happen to name a reference pair, and matched-edge counts this small make the macro-F1 move several points on a single label.

The two are kept apart from the metrics they resemble on purpose. Direction accuracy is not reference-edge recall, which requires direction to match before it counts an edge at all and so cannot say how often the sign was right on the pairs the model did find — which is exactly what makes B1-R's single reversal visible here and invisible there. Strength macro-F1 is not strength inflation, which counts only over-statement; macro-F1 is symmetric and includes under-statement, and macro-averaging stops a rare label such as `correlational` being drowned by `direct`. No condition was skipped for a cutoff mismatch, and the sensitivity arm is reported beside it in `reports/reasoning_diagnostics_sensitivity.md`.

Fixed-path label accuracy. Rebuilding three fixed synthetic paths from their edge labels reproduced weakest-hop strength and fully-sourced status on all three paths (3/3 for each). See `reports/path_label_check.md`. Strict path validity in Section 4.9 also checks the evidence and strength of the underlying edges on development outputs.

4.7 Structural outcome by condition

From a complete 42-graph sweep, seven conditions over six development candidates, logged and audited with zero failures. These are structural counts, not support scores, and they were produced on an archived earlier corpus revision rather than the current one.

Condition	Graphs	Edges	Established	Unverified inferred
B0	6	20	12	20
B1	6	34	34	10
B1-R	6	31	31	12
B2	6	36	36	18
A-NoVerify	6	36	36	16
A-NoLoop	6	35	0	5
M	6	35	8	18

4.8 Assistant-labelled development scores (10–11 November update)

Both current-v7, gold-backed development arms completed 28 graphs each and passed structural audit. The 24 prose corrections and 247 edge-support judgements behind the scores below were **produced by an AI assistant, not by an independent human reviewer**, and are reported as such; the provenance is set out below and in `reports/assistant_dev_support_review_20261110.md`, and a blind human sample over them is scored further down. These four-query values are **descriptive only**; the test half has not been scored. The primary Qwen3-8B arm remains primary under D-28.

Provenance of the support labels, stated plainly. The project's own contemporaneous record says the preparation and scoring scripts recorded all 24 correction graphs and all 247 support labels as assistant-produced drafts, that no blinding or independence was claimed when they were made, and that the official scorer twice refused them with *"complete independent blinded review attestation is required"*. The labels were subsequently rebound to fresh packets under a human attestation, with every label preserved unchanged. Preserving the labels and changing the attestation does not make them human judgements, so this report does not describe them as blind human review. The scores are arithmetically reproducible from the committed packets and were recomputed independently for this report; what is unestablished is not the arithmetic but the judgement underneath it.

How far the labels survive a human sample. A stratified blind sample of 40 of the 247 labels was drawn across arm and the assistant's own label, annotated by one author who did not hold the key and did not produce the labels, and scored with `scripts/sample_validate_support.py` (`reports/support_sample_validation.md`). Thirteen of the 40 rows carried no evidence span, so the only verdict available on the sheet was *unsupported* and agreement on them is agreement by construction; they are excluded from the figures below and reported separately.

measure, informative rows only	value
informative rows	27 of 40
claims the human rejected	14 of 27
raw agreement	0.778
chance agreement	0.495
Cohen's kappa	0.560 (95% CI 0.249 to 0.870)
disagreements	6 of 27

Every disagreement falls in the sensitivity arm. On the primary arm the human and the assistant agreed on all 10 informative rows; on the sensitivity arm they agreed on 11 of 17 (0.647). Because D-28 makes the primary arm the

source of every headline number, the figures in the table below rest on the better-agreeing half of the sample. That is the favourable reading and it is reported with its limit: ten rows cannot establish reliability. The key's label balance is now available: the primary informative sample contained six supported and four unsupported labels, and the human agreed on all ten. It is therefore not a constant-label comparison. That distinction is quantitative rather than rhetorical: a 6/4 split on both sides puts chance agreement at 0.52, so the primary arm's kappa is **1.000 against a real denominator**, where a dominant label would have driven chance agreement toward 1 and left the coefficient undefined — the failure mode the dev adjudication actually hit (Section 4.4). It remains a small stratified sample rather than proof that the primary labels are correct.

Arm	All drawn assistant yes / no	Informative assistant yes / no	Informative human yes / no
Primary	6 / 12	6 / 4	6 / 4
Sensitivity	11 / 11	11 / 6	7 / 10

The disagreements have a direction, and it matters for UER. Five of the six are the assistant crediting support that the human rejected; one is the reverse. An over-credited edge is counted as supported and so leaves the UER numerator, which means a label set biased this way *understates* UER. The sensitivity arm's UER could be understated if that pattern extends beyond the sample. This sample cannot establish a numerical correction or bound for either arm.

This sample measures one human against the assistant on 27 judgements. It does not convert the other labels into human ones, and every number in this section remains an assistant-labelled development diagnostic.

Arm	B1 overall UER	M overall UER	M–B1 paired UER	M–B1 reference recall	H2d
Primary Qwen3-8B	1.000 (3/4 queries)	0.750 (4/4)	−0.167 (3 usable pairs)	+0.062 (4 pairs)	holds descriptively
Sensitivity Qwen3.5-9B	0.529 (4/4)	0.250 (2/4)	−0.100 (2 usable pairs)	−0.062 (4 pairs)	holds descriptively

The primary B1 hop buckets have only 5, 2 and 1 established edges at hops 1, 2 and 3-or-deeper; the sensitivity B1 buckets have 13, 1 and 0. Neither arm clears the pre-declared three-bucket gate, so H1 is not testable on this dev set. H2b has zero usable paired queries in both arms for the change from hop 1 to hop 3 or deeper, for both UER and strength inflation. M's lower UER does not establish an inferential H2 result with four queries; reference-edge recall remains near zero. Missing denominators are reported as N/A rather than as zero error. Full per-hop strength inflation, recall, query counts and guard interpretation are in the RQ reports.

A predeclared one-query repeated-sampling check ran M on `dev-oil-effects` three times under the pinned primary model and retriever, changing only the sampling seed across repeats. The proposed link sets varied, but all three established the same two links with byte-identical evidence, matching the greedy run. Reusing the greedy run's assistant-produced support labels for those exact links gives UER 0.500 in each repeat. That reuse is an identity match on existing labels, not a fresh support review, so the figure inherits the provenance above — though these are primary-arm labels, the half of the sample on which the blind human check found no disagreement. This supports stability of the established set for **one query at three seeds**, not stability of the M-minus-B1 comparison across queries. Full counts and measured time are in `reports/stability_check_20261110.md`.

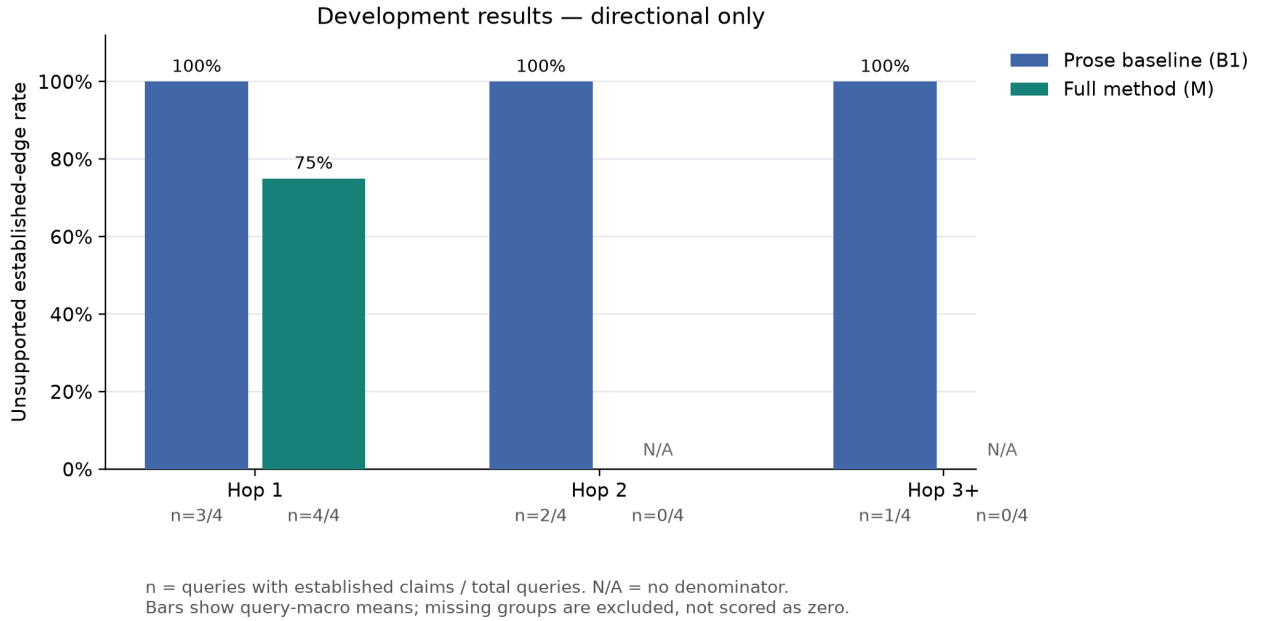


Figure 2. Assistant-labelled primary-arm UER by hop for B1 and M; labels show queries with denominators out of four.

4.9 Ablations, reranker baseline, and secondary metrics

The proposal's RQ2 analysis asks for four comparisons; until now only M against B1 had been computed. The other three are **exploratory** — the pre-declared tests are B1 deep-minus-shallow and M-minus-B1, and nothing here is promoted past that — but they are the comparisons that isolate each capability.

Contrast	Tests	Primary arm	Sensitivity arm
M – B2	**H2c**: beats prompted self-labelling	–0.250 (4 pairs)	–0.625 (2 pairs)
M – A-NoVerify	does verification help?	–0.208 (4 pairs)	–0.625 (2 pairs)
M – A-NoLoop	does targeted retrieval help?	+0.500 (1 pair)	n/a (0 pairs)

The UER component of H2c favours M descriptively in both arms. Its UER is lower than that of RAG prompted to label its own links, by 25 points in the primary arm and 62.5 in the sensitivity arm. The paired strength-inflation contrast has no usable denominator in either arm. The A-NoVerify contrast points the same way, which is the ablation that matters most for the central claim — removing the verifier makes the system worse by the measure the study is built on.

The A-NoLoop figure should not be read. It rests on a single query, because A-NoLoop established exactly one edge across the whole primary arm and none in the sensitivity arm. Its zero-width interval is one number resampled, not precision. Whether targeted retrieval helps is untested here.

Exploratory end-to-end reranker comparison. These are paired UER differences from the same assistant-labelled development scores. Negative values favour the first condition. The retrieval pilot is a separate comparison.

Contrast	Primary arm	Sensitivity arm
B1-R minus B1	–0.333 (3 pairs)	–0.092 (4 pairs)
M minus B1-R	0.000 (4 pairs)	–0.125 (2 pairs)

B1-R reduced UER relative to B1 in both development arms. M tied B1-R on primary UER and reference recall. In the sensitivity arm M's recall was 0.125 lower than B1-R's over four queries. The low pair counts prevent inference.

Two secondary metrics the proposal asks for had been computed by `eval.metrics` since Phase 2 and printed nowhere.

Condition	Provenance accuracy	Strict path validity
M	0.750	0.083
A-NoVerify	0.396	0.000
B1	0.750	0.000
B2	0.500	0.000
B1-R	0.750	0.125

Provenance accuracy is the metric that audits the instrument, since the provenance label is the mechanism the whole study measures with. M reaches 0.750 against A-NoVerify's 0.396 in the primary arm, and 0.917 against 0.104 in the sensitivity arm: the verifier roughly doubles the rate at which an edge's source-asserted or model-inferred label is right, which is a more direct defence of it than the UER difference.

Strict path validity is near zero almost everywhere — 0.083 for M, 0.000 for most conditions, never above 0.125 in the primary arm. A path counts only if every hop is established, supported and no stronger than its source, so a chain is as weak as its weakest link by construction. Almost no multi-hop explanation in this corpus survives that test end to end. That is a finding about the difficulty of the task rather than a defect in any one condition, and it is the hop-by-hop claim of the study restated at the level of whole paths.

All of these inherit the label provenance set out in Section 4.8 and are development figures on four queries.

5. Discussion, Failure Analysis and Mitigation

5.1 Where deterministic checking ends and judgement begins

The challenge-set result is the clearest evidence the project has, and it points in both directions at once. Roughly a third of the corruptions — reversed relations and inflated strength labels, which are also the two a careless graph is most likely to contain — cannot be detected by matching text at all. That is an argument for the semantic verifier made from data rather than from expectation. Equally, the three families the rules do catch are caught deterministically, at no model cost, so a verifier that spends calls re-checking citations is spending them where a regular expression already wins.

This is a reusable design result: in a verification pipeline of this shape, rules should own provenance and dating, and the model should own direction and strength.

5.2 Does verification repair, or does it delete?

The ablations make the mechanism visible. Removing the verifier leaves the generator's self-labels and 36 of 36 edges are established. Keeping the verifier but removing the search loop leaves 0 of 35. The full system establishes 8 of 35.

Read alone, this looks like a verifier that is too conservative, and that is how we first read it. The model pilot in Section 4.1 suggests a different reading. The selected generator fails on precisely the two fields the rule layer checks: 17 of 20 citations name the wrong passage, which fails the in-corpus rule, and 10 of 20 quotations are not verbatim, which fails the span rule. The rejected alternative model had **zero** failures in both categories.

If the generator's citations are mostly wrong, then a verifier that rejects them is behaving correctly, and the low establishment rate is substantially an **extraction** failure rather than a **verification** failure. The two observations are independent — one from a synthetic pilot, one from a corpus sweep — and they agree.

This has a direct consequence for the selection in 4.1. The rule averaged five tasks unweighted and tied at exactly 61.0%, so latency chose the model. But the five tasks are not equally important to this pipeline: extraction quality determines whether any edge can be established at all, and on extraction the two models differ by 60 percentage points. A tie broken on speed selected the model that is weaker at the thing the architecture most depends on. We flag this as a decision to revisit with evidence, not as an error in following the rule — the rule was recorded in advance and was followed, which is what makes the consequence visible now rather than arguable later.

5.3 What the design cost us

Reality reduced the study. Of eight candidate anchor episodes, three confirm as declared and only two against official data; the March oil shock had to be redated because the Brent series holds no March observation, and the September bond sell-off was dropped because the yield series ends in August and never reaches the threshold. That reduced 30 planned queries to 8, which in turn caps the signed gold set at 24 edges. The challenge set exceeds the planned total, with two fewer clean controls than planned.

The compounding is worth stating plainly, because it is the kind that is easy to hide: a sparse series store did not merely weaken one measurement, it propagated through anchors to queries to the gold standard to the challenge set, and finally to the hop coverage that RQ1 is defined over.

5.4 Qualitative failures and mitigation

Observed development case	Failure	Mitigation tested or planned
A-NoVerify, rupee effects	One paragraph was reused for six established claims it did not state	Semantic verification tests the relation after source eligibility
A-NoVerify, rupee why	A cited oil-to-rupee relation was reversed, and one quote was nonverbatim	Verbatim checking filters the quote. Direction verification is measured on the challenge set
B2, oil why	Six established claims had no attached evidence	M requires verified support to mark a link sourced. The recall guard records how much is lost
M, both arms	Four of four queries exhausted the 12-call budget with open edges	Report remaining edges and defer testing a revised repair policy to a new run

These examples come from the saved development artifacts. The first three use the same assistant-labelled judgments as Section 4.8. The five-case ranking and artifact checksums are in `reports/development_failure_analysis_v7.md`.

6. Limitations

Held-out support scores pending. The primary model completed all 14 held-out runs across seven conditions and two queries about one FPI episode. Section 4 reports descriptive development UER and associated diagnostics. The held-out sample falls below the ten-query inference gate. Its references also seed some items in the separate verifier challenge diagnostic. A-NoVerify and B2 each produced one nonverbatim span in the held-out sweep.

The gold set reaches hop 2. Twelve of 16 gold edges are at hop 1 and four at hop 2, with none deeper. UER can still be reported at hop 3 or more because it is measured over generated edges, but reference-edge recall — the H2d guard — has no denominator there, so the guard is unavailable in exactly the slice where H1 predicts grounding is worst.

Runtime confirmation. The offline episode audit found three confirmed episodes as declared, including two against official data. The scored query inputs did not declare runtime observation windows. All 56 saved development runs therefore recorded an unconfirmed anchor. Their as-of evidence filters still ran. These are development analyses of declared events. The separate versioned query manifests define explicit thresholds. The held-out sweep used those thresholds and confirmed both test anchors in every condition.

Source concentration. 870 of 1,095 news documents, 79%, come from one publisher. Business Standard and the RBI refuse `robots.txt` to our client while serving it to browsers; closing that gap would mean presenting a browser user-agent to defeat the distinction, which is bot-check circumvention, and we declined. Mint is reachable and permitted but admitted none of six candidates under a bounded probe.

Alias coverage caps reference-edge recall before any model runs. Rule 3 requires both variables of an edge to be named in the cited passage, through a hand-written alias table. Auditing that table against all 16,472 passages (`reports/alias_coverage.md`), **three of the 46 ontology variables are never named at all** — `AIRLINE_SHARES`, `CORPORATE_BOND_SPREAD` and `CPI_FUEL_LIGHT` — and five more appear in fewer than ten passages. A variable the corpus never names cannot carry a source-asserted edge however much the sources discuss the underlying quantity.

This has a specific and uncomfortable consequence. The proposal's own illustrative chain runs crude → ATF price → airline costs → airline shares, and its last two links are exactly the unevidenceable ones: `AIRLINE_COSTS` appears in two passages and `AIRLINE_SHARES` in none. **The worked example used to motivate the study cannot be built from the corpus the study froze.** It is also a deep-hop chain, so this compounds the hop-2 ceiling above rather than being separate from it.

Two causes are possible and they need opposite fixes: either the corpus does not discuss these quantities, in which case queries touching them must be rescoped, or the sources use wording the alias table lacks, in which case the table is incomplete. Distinguishing them requires reading the corpus, and that reading has now been done (`reports/alias_review_20261110.md`). It separates cleanly. **Two** forms are genuine wording gaps: "IndiGo and SpiceJet shares" names `AIRLINE_SHARES` and "fuel component of CPI" names `CPI_FUEL_LIGHT`, both in Indian coverage the table fails to match. The remaining candidates are corpus gaps or traps — "fuel inflation" and "airline stocks" also occur in generic and European coverage where a substring alias would overmatch, "InterGlobe Aviation" can name a company or a financing rather than a share, and every corporate-bond-spread candidate turned out to be a sovereign or repo spread.

We have still not extended the table, and the reason is no longer uncertainty (D-29). The table is an input to rule 3, so changing it changes the verifier for every condition at once; the 56 development graphs, their 247 support judgements, the challenge set and every score in Section 4 were produced under the pinned table. Applying two forms now would require a fresh versioned sweep and a fresh review of every edge whose eligibility moved, and would leave these numbers describing a configuration that no longer exists. So the ceiling is reported as what it now is: a **located** bound with two named repairs outstanding, rather than an unexplained floor.

The reviewer evidence graphs are small. The 16 edges are span-exact, adjudicated by two independent annotators and covered by a signed review receipt bound to their bytes (Section 4.4), so the reference set is no longer provisional. It is, however, exactly at the declared four-edge-per-graph minimum, so no further edge can be dropped without re-drafting, and 16 edges is a thin basis for any agreement coefficient — which is why Section 4.4 reports the raw agreement beside the kappa rather than the kappa alone.

Pilot scale. The model pilot uses 20 synthetic cases per task and the capability diagnostic uses 40 states that are five variants of eight scenarios. Both are development diagnostics, not benchmarks.

Other review limits. The earlier relevance-judgement comparison and safety screen have their own provenance and limits (Sections 4.2 and 4.6). The 10 November prose and edge-support labelling was assistant-produced; its provenance, and the competing attestations around it, are set out in Section 4.8 and `reports/assistant_dev_support_review_20261110.md`.

Schedule. The blind double annotation planned for 30 October took place on 8 November, nine days late. It was the first step of the dependency chain in Section 7 and the only one with no slack, so the sweep and every number measured against the gold set moved with it.

7. Remaining work

The final submission is 15 November. The milestones below are current as of 12 November.

Status	Work
Done, 8 Nov	Blind double annotation of the 20 drafted development gold edges; agreement scored; four edges dropped in reconciliation, leaving 16 (Section 4.4, D-26)
Done, 8 Nov	Reference review receipt signed and hash-bound, which makes the challenge set's false-alarm rate quotable (Section 4.5)
Done, 8–9 Nov	Pilot decisions D-01 to D-04 locked; the two-model sensitivity check pre-registered (D-28)
Done, 10 Nov	Both gold-backed development arms: 56 graphs, two complete audits
Done, 10 Nov	Both sweep arms run and scored; 247 assistant-produced support labels and 24 prose corrections, reported as such (Section 4.8)
Done, 10 Nov	Stratified blind human sample over those labels: 27 informative rows, raw agreement 0.778, kappa 0.560, every disagreement in the sensitivity arm (Section 4.8)
Done, 10 Nov	Development UER, strength inflation, reference recall, sensitivity comparison and UER figure, with denominator gates
Done, 10 Nov	Five-case primary-arm failure analysis; earlier clean-clone reproduction remains documented
Done, 11 Nov	Repeated-sampling stability check: three seeds on `dev-oil-effects` under M; the established set and its UER of 0.500 identical across repeats, the proposal stage visibly stochastic (`reports/stability_check_20261110.md`)
Done, 11 Nov	Test gold drafted for two queries, then **adjudicated**: both annotators agreed on every surviving edge and every direction, split on seven of eight strengths, reconciled under D-30; receipt signed (D-27a, D-30)
Done, 12 Nov	Signed-reference retrieval diagnostic on the two FPI queries, with and without the reranker
Done, 12 Nov	Controller metrics restored, fixed-path check added, and reproduction checks rerun
Done, 12 Nov	Public-data model benchmark completed on 400 frozen items
Done, 12 Nov	Primary held-out generation on both signed FPI queries in all seven conditions. The audit verified 14 complete, checksum-bound artifacts
Pending	Full human prose correction and independent blind support judgments for the development and held-out outputs
Pending	Expansion to the original signed test coverage. Fourteen candidate questions were screened, with one unsigned reference draft prepared for independent review

The signed test set has two queries. Reference graphs exist for `test-fpi-why` and `test-fpi-effects`, drafted from the corpus with every span sliced verbatim and passed through the rule layer, then adjudicated — four edges each after reconciliation, across hops 1 and 2. The adjudication itself is described below. The third query the feasibility audit called feasible did not survive reading. The audit counted variables co-occurring with

CPI_HEADLINE and said plainly that it was counting co-occurrence; the passages show the condition failing structurally, because in this corpus Indian headline CPI is the *destination* of the oil shock rather than its source. BRENT, FUEL_PRICES, CPI_FOOD, CPI_CORE and WPI all run into CPI, and an effects graph rooted at CPI cannot use an inbound edge; what points outward is CPI to the repo rate, conditionally, and little else (D-27a). Two queries sit five times below MIN_QUERIES_FOR_INFERENCE. The signed graphs are eligible references for a descriptive held-out comparison once the output review is complete.

The query the audit also ruled out. A reachability audit run before any drafting (reports/test_gold_feasibility.md) had already excluded test-cpi-why on cheaper evidence. That query asks why inflation was *raising expectations of RBI tightening*, so its graph must reach RATE_HIKE_EXPECTATIONS, and on or before its as-of date exactly one passage in the corpus pairs that variable with headline CPI. That passage belongs to a Federal Reserve article rather than Indian coverage, and it runs the wrong way — markets reducing rate-hike expectations after a *softer* CPI print, against an anchor of CPI rising. Widening the alias table would not change that, because the variable is named in 33 passages and simply never beside headline CPI in Indian coverage. The two exclusions were found by different means, and the order matters: the audit is cheap and settled one query before anyone read a passage, while the other needed reading. Reporting both is what makes the test half's size a measurement rather than a shortfall.

The test half is adjudicated, and it agreed where it counts. Both annotators labelled all ten drafted edges independently and blind. They agreed on asserts at raw 1.000 (kappa 1.000) and on direction at raw 1.000 — kappa undefined there, because one label was used throughout, which is a property of the sample rather than a failure to agree. They split on **seven of eight strengths** (raw 0.125, kappa 0.000), and the pre-declared direct-versus-hedged collapse did not rescue it, because that is precisely the axis of the split (reports/adjudication_kappa_test.md). Reconciliation applied one rule, D-30: strength is read off the modality of the clause carrying the causal claim, not off the manner in which the mechanism is described, so a claim under "would" is conditional however firmly the mechanism inside it is worded. Both annotators had independently rejected the same two FPI_FLOWS -> GSEC_10Y edges, which are dropped, leaving **four edges per graph at exactly** MIN_EDGES. The receipt is signed and hash-bound to those bytes.

The prototype's interactive output exists. The proposal's Section 8 ends by asking for "an interactive event-impact graph as its final output", and it was dropped on 29 October as unreachable alongside the experiments. The development experiments finished, so it was built: reports/graph_explorer.html, one self-contained file with no network dependency, where every edge opens to the span it rests on with its passage id, publication date and the cutoff it had to clear — or, for a model_inferred edge, to the statement that no verified source was found. It is opened from disk rather than served, because the spans are corpus excerpts under a research-use licence. The viewer also includes a synthetic graph with inferred edges and path labels. A local export displays the full-system graph from the one-query smoke run. The viewer lets a reader do what Section 3 describes a reader wanting to do, which is follow a chain and see where it stops being grounded.

Working demos. The repository README links to a downloadable synthetic browser demo and its runnable seven-condition pipeline. It also provides a local-model command using the frozen assets and a dated dev query. The 12 November live execution completed all seven conditions with bound logs and confirmed anchors. No retrieved passage exceeded its cutoff. B1-R produced two nonverbatim spans on this query. Those baseline failures remain visible and this demo adds no human-scored result. Demo and reproduction instructions

The generation audit records completion, confirmed anchors and structural checks. Human review starts with six held-out and 24 development prose outputs. Corrected graphs then go into condition-blind support packets. Fourteen expansion candidates across seven episodes await independent reference review. Held-out generation audit

The model-selection sensitivity check addresses the concern raised in Section 5.2: the D-01 selection is re-examined not by argument but by running the sweep on both models under a comparison declared in advance (D-28, docs/preregistration_model_sensitivity.md). The selected model remains primary whatever the two arms show; the second arm exists to say whether any rise in UER with hop survives a generator that does not fail at citation

mechanics.

8. Conclusions

We have built and tested the apparatus for measuring how evidential support and causal strength change hop by hop in LLM explanations of real financial news. Both development arms have scored support labels, though those labels are assistant-produced rather than independently human-reviewed, and are reported as such. A blind human sample of 40 of them agrees with the assistant on 0.778 of the 27 informative rows (kappa 0.560), with every disagreement falling in the sensitivity arm and five of six in the direction that flatters the labels — so the sensitivity arm's error rates read as a floor. On them, the primary arm's M-minus-B1 UER difference is −0.167 over three usable query pairs and the sensitivity arm's is −0.100 over two. Neither arm meets the predeclared hop-bucket gate for an H1 trend claim, and four development queries cannot support inference. The two-query held-out sweep generated all 14 outputs and awaits independent support scoring.

What the work already shows is that the verification question is sharper than we posed it. Deterministic rules and semantic judgement divide cleanly and measurably, with rules owning provenance and the model owning meaning — and the challenge set now puts a number on the second half. On the 38 corruptions no rule can see, because a reversed edge keeps the same variables, passage and span, the rule layer catches 0 and the verifier catches 17, for 2 false alarms in 23 clean controls (Section 4.5). Verification is therefore worth having and is not close to sufficient, which is a more useful finding than either half alone. And the striking scarcity of established edges, which we first attributed to an over-cautious verifier, appears instead to trace to a generator that cites the wrong passage in 17 of 20 cases — which is a different problem, with a different fix, and one we would not have seen without running both diagnostics.

References

Asai et al. (2024). Self-RAG. ICLR.

Chen et al. (2024). BGE M3-Embedding. arXiv:2402.03216.

Chi et al. (2026). WorldReasoner. arXiv:2606.11816.

Cormack, Clarke and Büttcher (2009). Reciprocal rank fusion. SIGIR.

Feinstein and Cicchetti (1990). High agreement but low kappa: I. The problems of two paradoxes. Journal of Clinical Epidemiology 43(6).

Gendron et al. (2025). Causal Cartographer. arXiv:2505.14396.

Mariko et al. (2020). FinCausal 2020. FNP.

Prasad et al. (2023). ReCEval. EMNLP.

Qwen Team (2025). Qwen3 technical report. arXiv:2505.09388.

Romanou et al. (2023). CRAB. EMNLP.

Tan et al. (2022). The Causal News Corpus. LREC.

Wang et al. (2025). CausalRAG. Findings of ACL.

Xie et al. (2025). FIRE. Findings of NAACL.

Yu, Li and Wang (2019). Detecting causal language use in science findings. EMNLP-IJCNLP.